

Microwave Engineering

Branch: ECE (4th Sem)

UNIT-5

MICROWAVE MEASUREMENT

- The RF/microwave engineer faces unique measurement challenges.
- At high frequencies, voltages and currents vary too rapidly for conventional electronic measurement equipment.
- Conventional curve tracers and oscilloscopes are of limited value when microwave component measurements are needed.
- In addition, calibration of conventional characterization equipment typically requires the use of open and short circuit standards that are not useful to the microwave engineer.
- For these reasons, most commonly exploited microwave measurements focus on the measurement of power and phase in the frequency domain as opposed to voltages and currents in the time domain.

- At audio frequency and low radio frequency range the direct measurement of voltage and current is possible.
- At microwave frequencies voltages, current and impedance are difficult to determine.
- Their values may vary at different points in the circuit and they are affected significantly by small changes in geometry.
- In the microwave frequency region direct measurement of power is more accurate.
- To measure the impedance of a load, reflection coefficient is measured for a known characteristics impedance of the transmission line.
- Methods of frequency measurement are also important because various characteristics of microwave components and networks are frequency dependent.

- At lower frequencies, direct measurement of y-, z-, or h-parameters is useful to accomplish linear characterization.
- However, RF and microwave design utilizes S-parameters for this application.
- Other small signal characteristics of interest in microwave design include impedance, VSWR, gain, and attenuation.
- Each of these quantities can be computed from two-port S-parameter data.
- The S-parameters are complex quantities normally expressed as magnitude and phase.
- Notice that S_{11} and S_{22} can be thought of as complex reflection ratios since they represent the magnitude and phase of waves reflected from port 1 (input) and 2 (output), respectively.

- It is common to measure the quality of the match between components using the reflection coefficient defined as

Input Reflection Coefficient, $\Gamma_{in} = |S_{11}|$

Output Reflection Coefficient, $\Gamma_{out} = |S_{22}|$

Reflection coefficient measurements are often expressed in dB and referred to as return loss evaluated as,

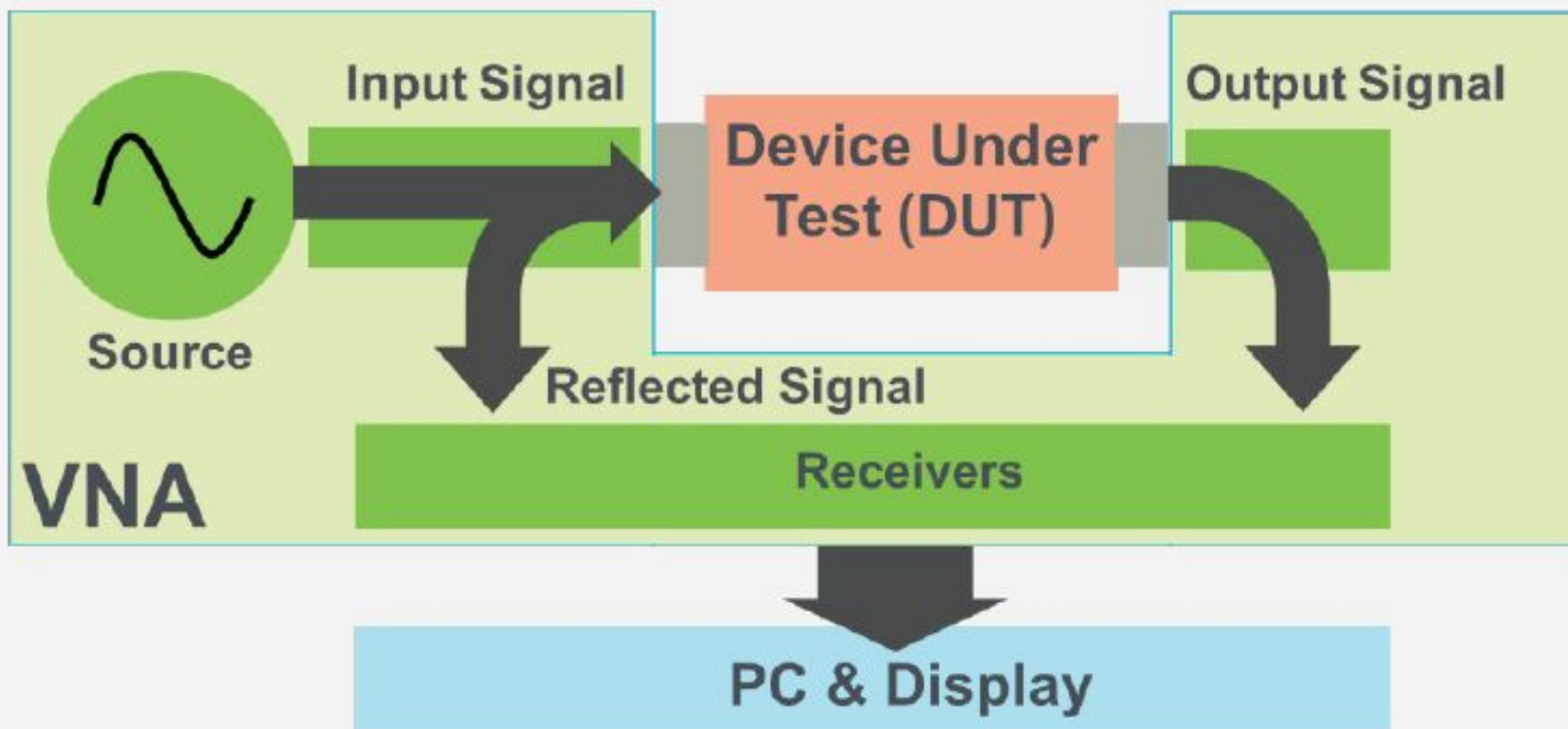
$$RL = -10 \log |\Gamma|^2 = -20 \log |\Gamma|$$

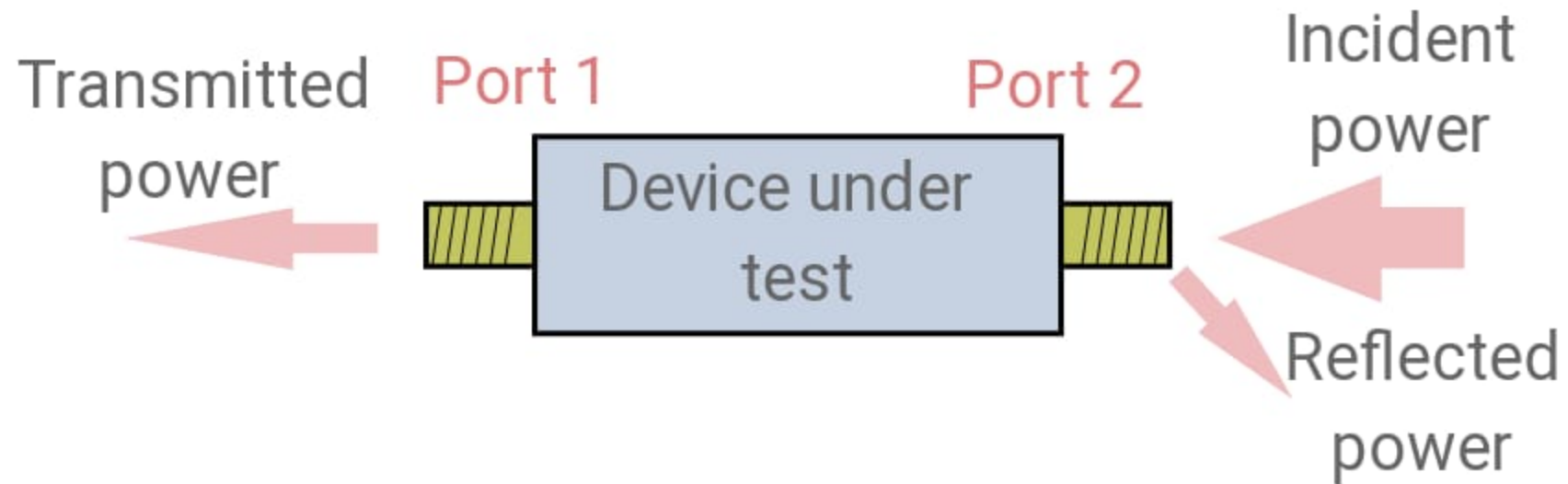
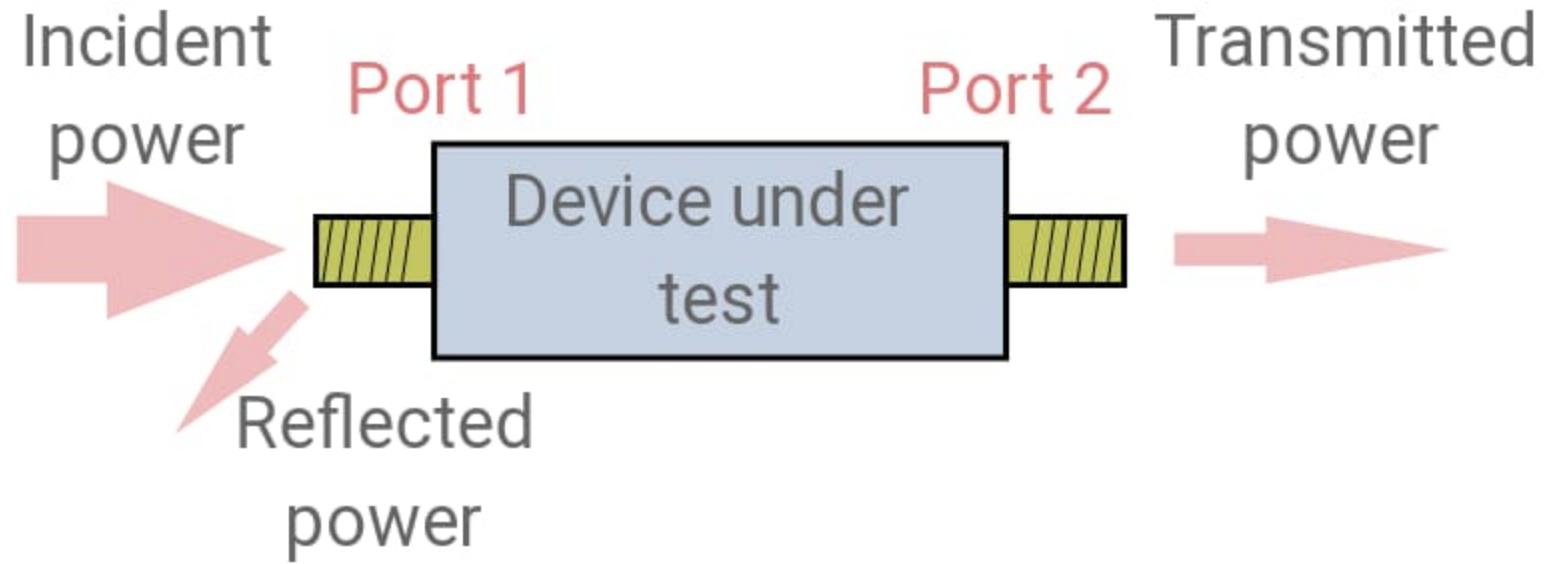
Forward Transmission Coefficient, $T_F = |S_{21}|$

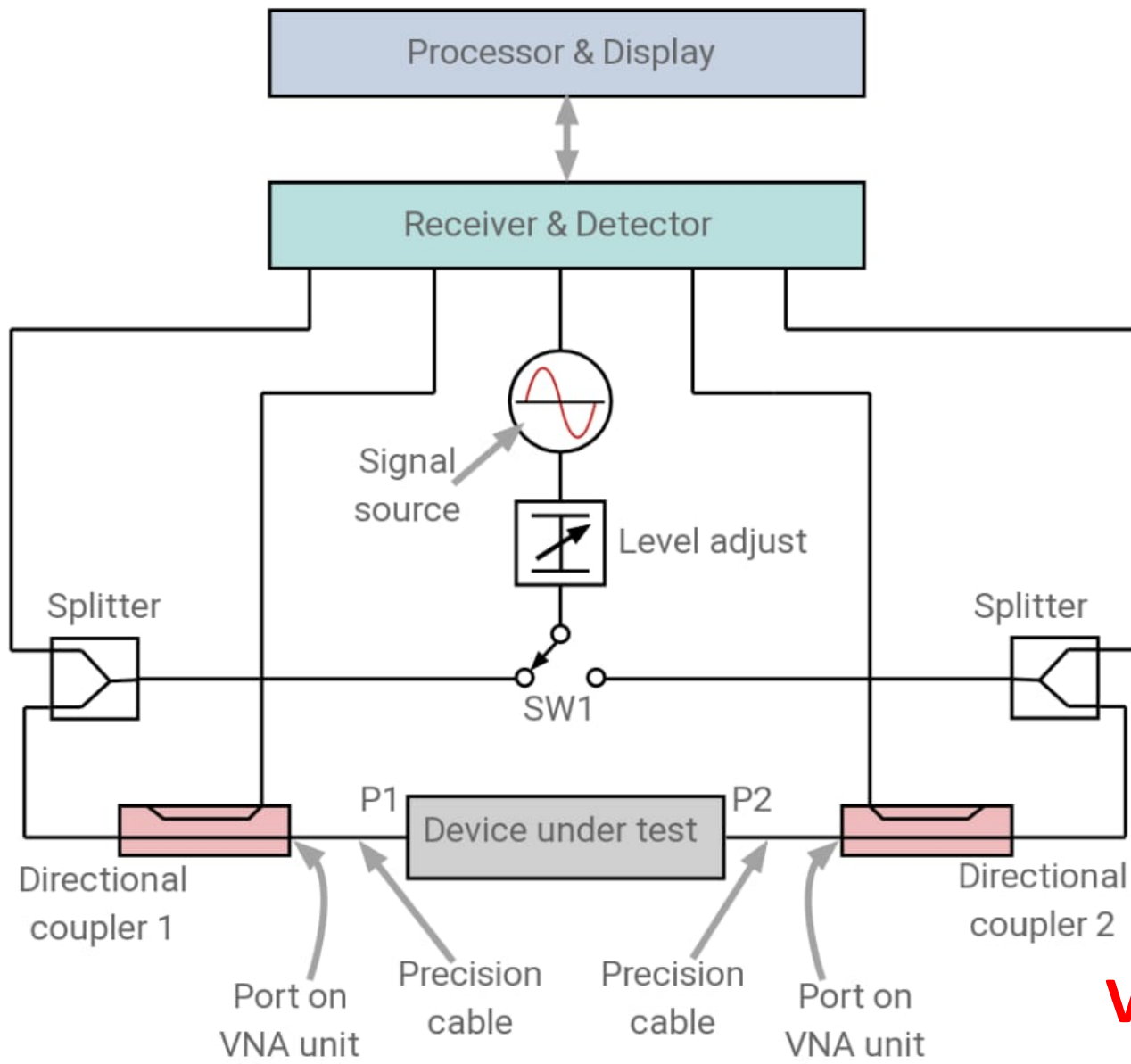
Reverse Transmission Coefficient, $T_R = |S_{12}|$

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

VECTOR NETWORK ANALYZER (VNA)







VNA internal circuit



Anritsu Advancing beyond

India



PRODUCTS

TECHNOLOGIES

INDUSTRIES

BUY

SUPPORT



MS2036C
VNA Master + Spectrum
Analyzer

VNA: 5 kHz — 6 GHz
SPA: 9 kHz — 9 GHz

≥ 90 dB (Transmission @ 6 GHz)

350
μsec/point

≥ 42 dB



MS2037C
VNA Master + Spectrum
Analyzer

VNA: 5 kHz—15 GHz
SPA: 9 kHz—15 GHz

≥ 85 dB (Transmission @ 15
GHz)

350
μsec/point

≥ 42 dB



MS2038C
VNA Master + Spectrum
Analyzer

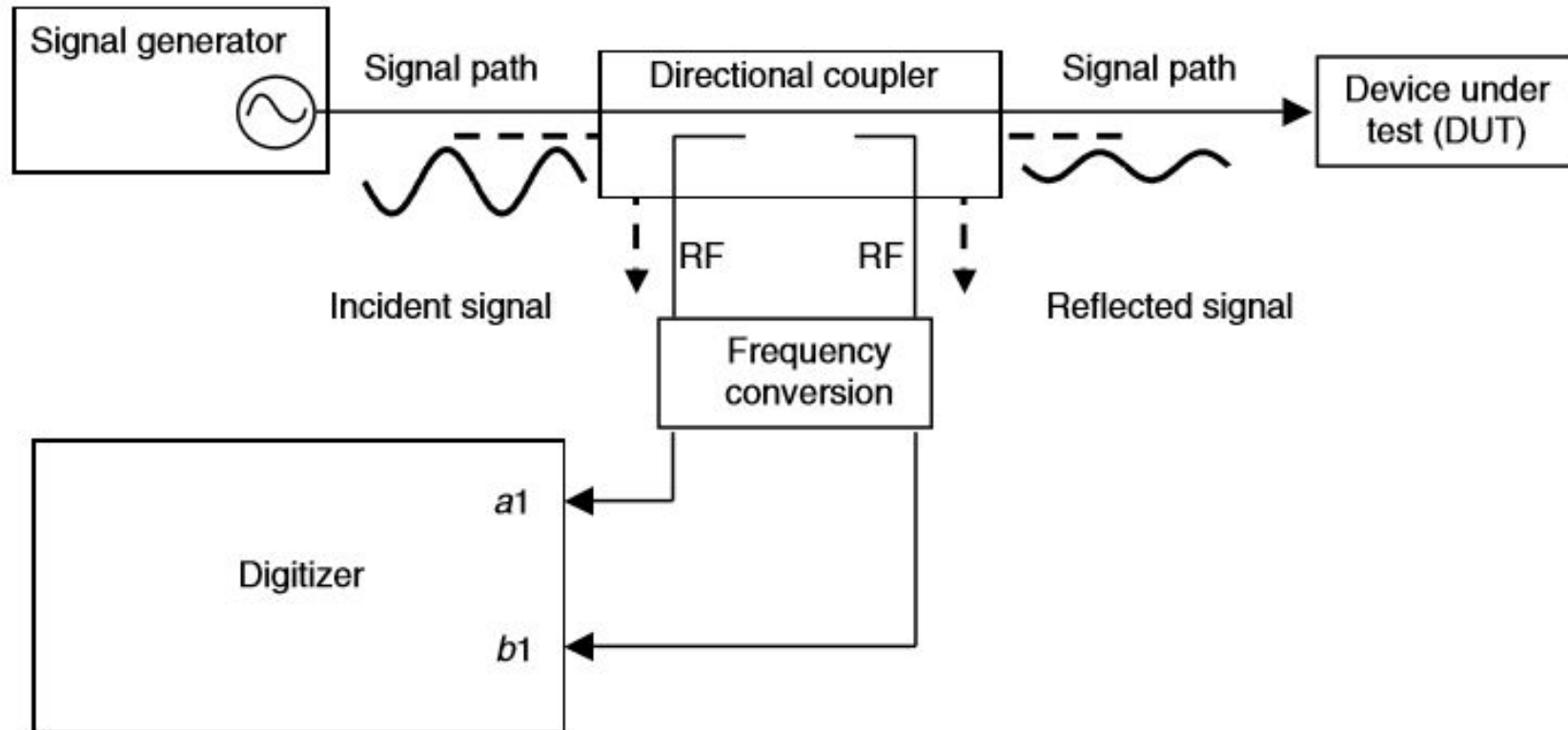
VNA: 5 kHz — 20 GHz
SPA: 9 kHz — 20 GHz

≥ 85 dB (Transmission @ 20
GHz)

350
μsec/point

≥ 42 dB

Practical VNA



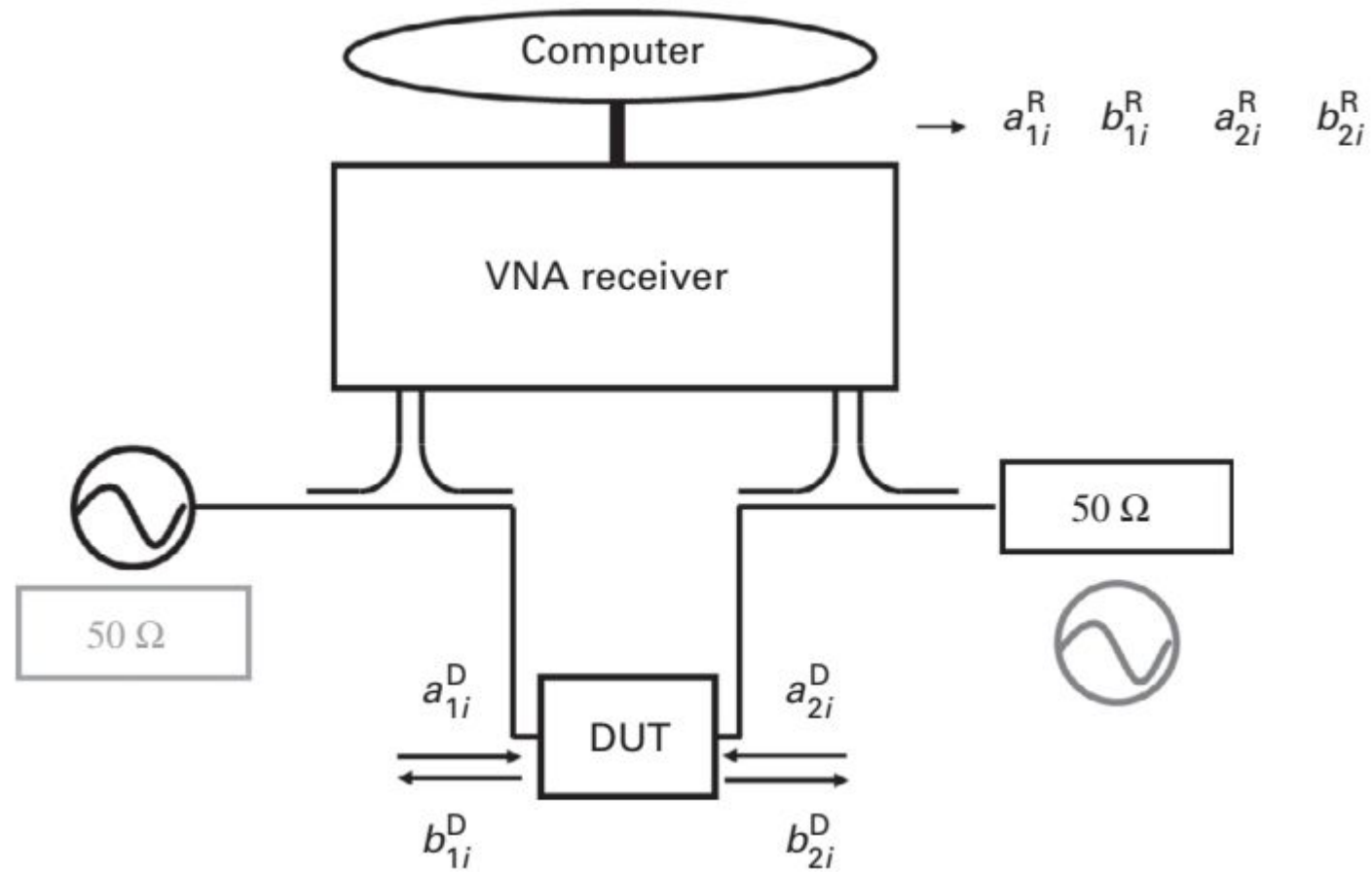
Vector network analyzer measurement configuration to determine S-parameters of a high-frequency device, component, or subsystem.

Working of VNA

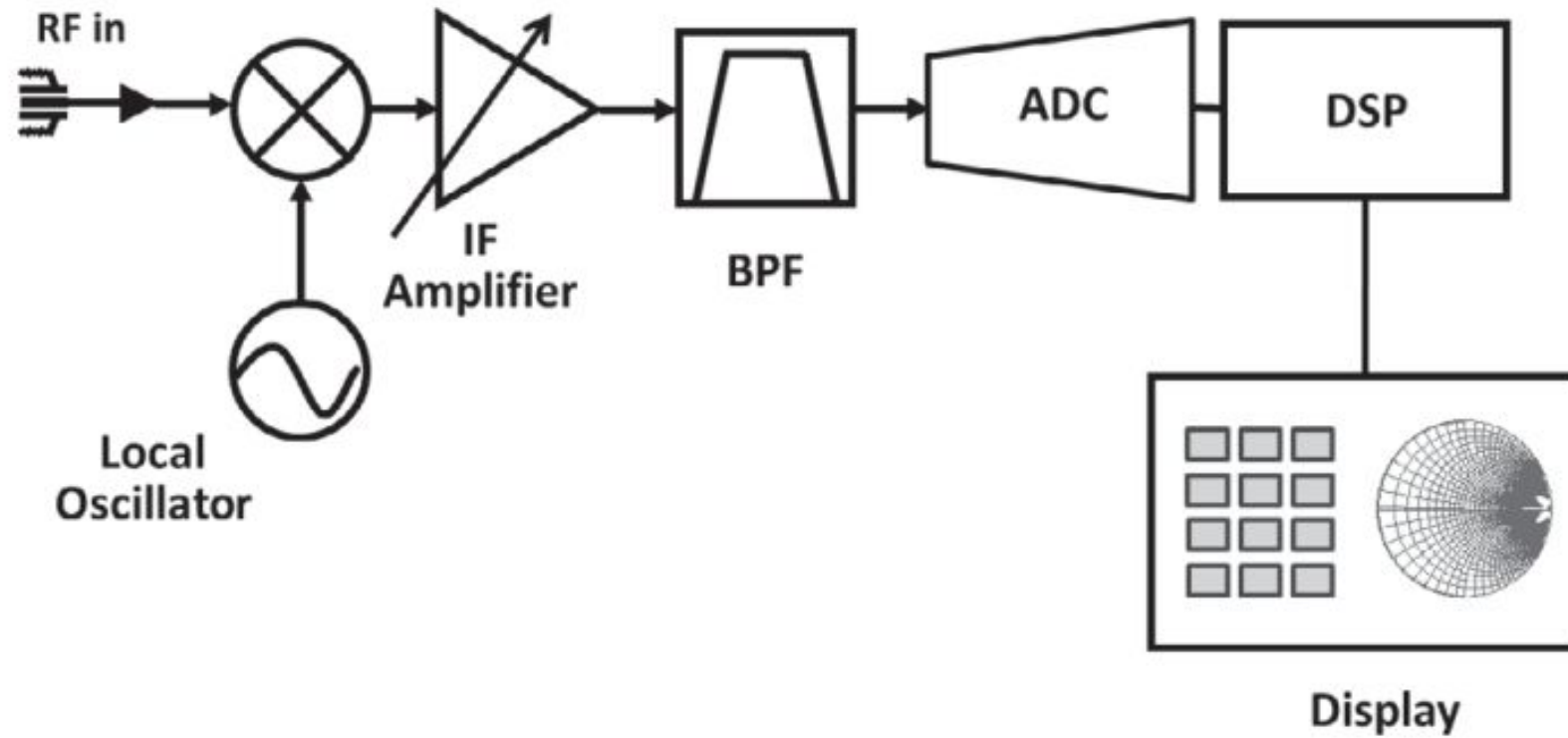
- The vector network analyzer (VNA) is the instrument of choice for small signal characterization of high-frequency components.
- Uses a source with well-defined impedance equal to the system impedance and all ports of the device under test (DUT) are terminated with the same impedance.
- During the network analyzer measurement, a signal is driven through the directional coupler to one port of the DUT. Part of the incident signal is sampled by the directional coupler.
- On arrival at the DUT port being measured, some of the incident signal will be reflected. This reflection is again sampled by the directional coupler. The sampled incident and reflected signals are then down converted in frequency and digitized.
- The measurement configuration shows only one-half of the equipment required to make full two-port S-parameter measurements. The S-parameters are determined by analyzing the ratios of the digitized signal data.

Vector Network Analyzer





Fundamental blocks of VNA

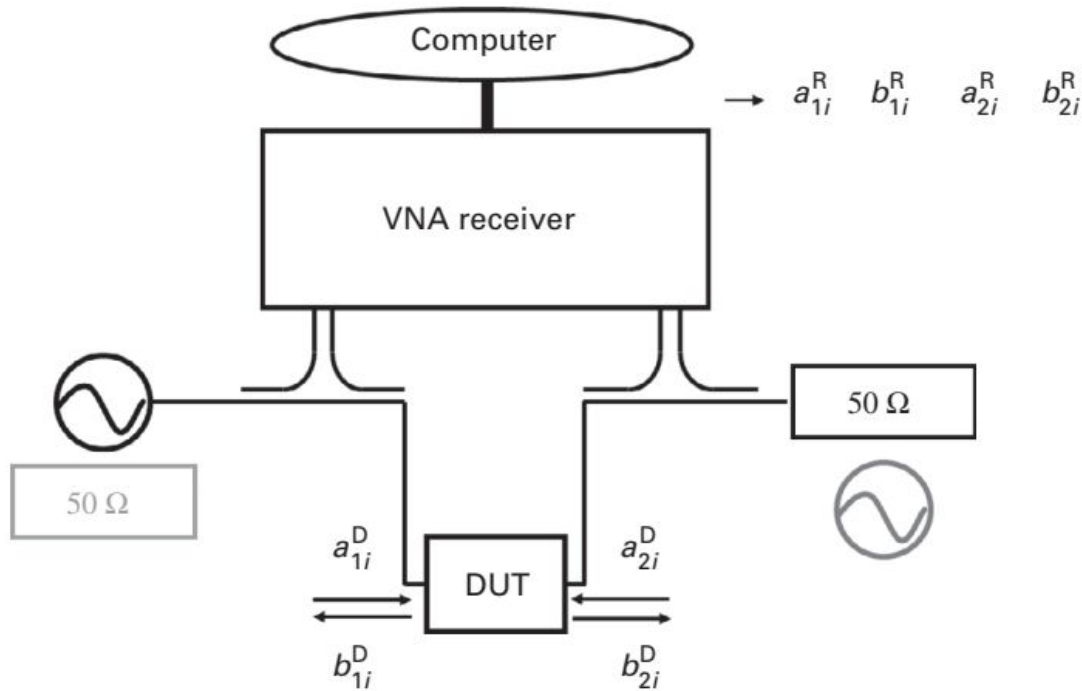


The receiver section of Vector Network Analyzer

Receiver Blocks

- The configuration has three blocks: a mixer, a filter, and an analog-to-digital converter.
- The input signal is down-converted to a lower intermediate frequency (IF) by mixing the input signal with the local oscillator (LO). The LO is locked to either the RF or the IF signal so that the receivers in the network analyzer are always tuned to the RF signal present at the input.
- The IF signal is bandpass filtered, mainly to remove unwanted signals created in the mixer, thereby improving the sensitivity and dynamic range. Note that, in the case of the VNA, a two-receiver configuration is needed so that one can measure the amplitude and phase of a scattered wave relative to an incident wave, or, in other words, obtain the S -parameters.
- After filtering, the amplitude can be measured using an envelope detector, and the phase may be measured with a quadrature detector. However, modern network analyzers use an analog-to-digital converter (ADC) and digital signal processing (DSP) to extract magnitude and phase information from the IF signal.
- Finally, the measured S -parameters are displayed in a format that can be interpreted, such as a Smith-chart plot. For the overall configuration to work properly, the instrument should be carefully calibrated prior to performing any measurement.

Calibration of VNA



- The quantities at the DUT's ports, indicated by superscript D and for the instrument it is represented by the raw quantities, indicated by superscript R.
- The aim of the process of calibration is to remove the systematic errors between the connector ports (connectorized measurement), or probe tips in the case of on-wafer measurements, and the actual data acquisition. This includes effects such as losses and phase shifts in cables, imperfect directional couplers, mismatches, etc.
- In general terms, the relationship between the four "DUT" waves and the four "raw" waves can be expressed by a 4×4 matrix.

The aim of the calibration process is to determine the unknown matrix coefficients.

Calibration Techniques

There exist many calibration procedures for S-parameter measurements.

The choice is essentially determined by the validity of the underlying assumptions.

- Most important technique is SOLT calibration, which is the most commonly adopted approach for connectorized measurements in the frequency range for wireless applications.
- TRL calibration is useful when one wants to develop one's own calibration kit.

SOLT Calibration



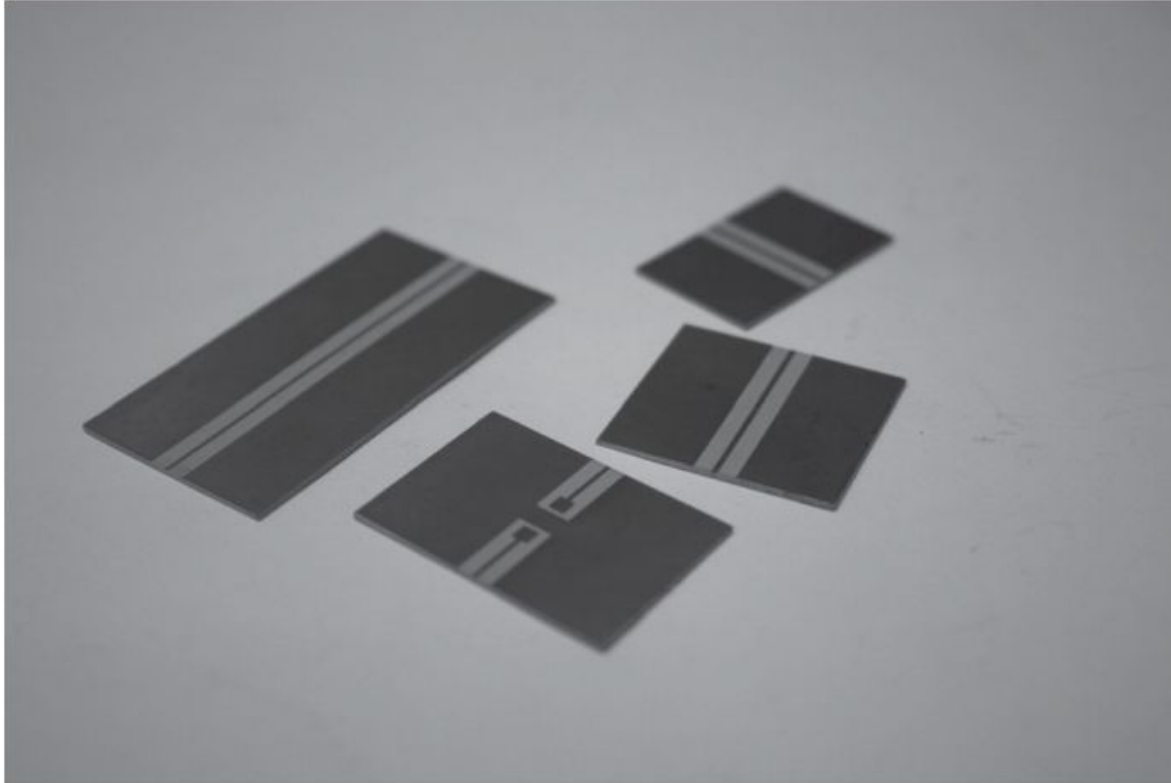
Agilent Calibration Box

This calibration involves a small box that the user has to connect only once, and then the various standards are applied sequentially by means of internal switches.

- Each standards in SOLT are not perfect, and therefore the manufacturer supplies their actual reflection coefficient .
- For instance, the “open” standard has a parasitic capacitance value, and the “load” has a residual inductance value due to the short transmission line connecting the connector to the internal 50 ohm resistor.
- This reflection coefficient is frequency dependent.

Short Open Load Through

TRL Calibration



- It is an alternative technique to SOLT, especially when the instrument manufacturer's connectorized calkit or, in the case of on-wafer measurements, calibration substrate is not applicable for the device to be measured.
- SOLT requires an accurately trimmed load. This is a challenge when one has to develop one's own calibration kit. TRL is less stringent fabrication wise.
- It does not require a load, the reflect doesn't have to be accurately known, and the characteristic impedance Z_0 of the through and line should be equal, but the value does not have to be known.

TRL calibration kit for in-fixture measurement

Requirements for TRL standards

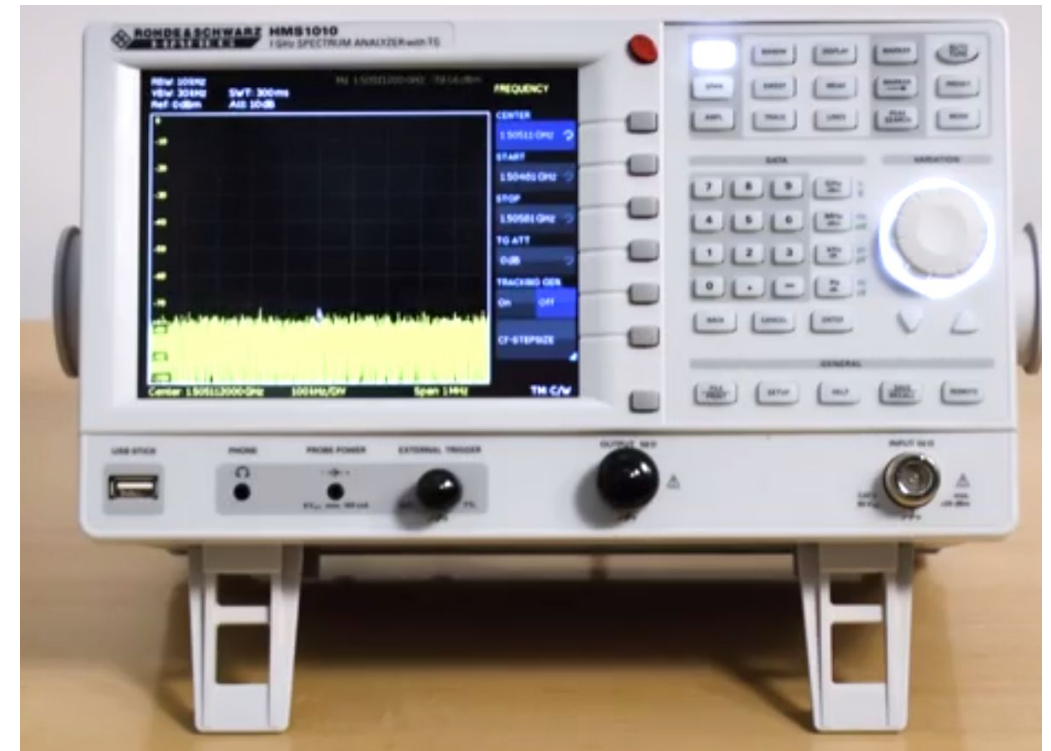
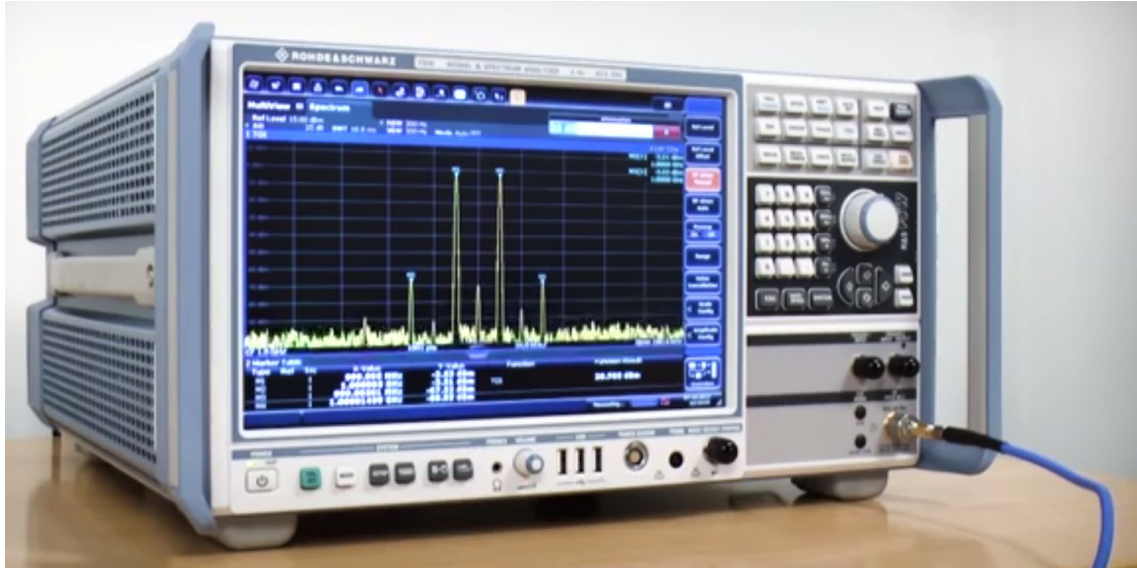
- Through Z_0 should be the same as Z_0 of the line standard. The attenuation doesn't have to be known. If it is a non-zero through, the electrical length should be well known and specified if the through is going to be used to set the reference plane.
- Reflect the magnitude of the reflection coefficient is optimally 1.0, but it is not necessary to know the exact value. The phase of the reflection coefficient must be known to within 90° accuracy. However, if the reflect is used to set the reference plane, its phase response must be well known and specified. Also, the reflect must be identical on both ports. Typically, one realizes either a short circuit or an open circuit.
- Line the characteristic impedance Z_0 of the line establishes the impedance of the measurement (i.e., $S_{11} = S_{22} = 0$). The insertion phase of the line must be different from that of the through. The electrical length difference between the through and the line must be larger than 20° and smaller than 160° . The attenuation doesn't need to be known. The electrical length should be known.
- Owing to the requirement that the electrical length difference between the through and the line must be between 20° and 160° , the bandwidth of the calibration might not be sufficient for the targeted measurements. In such cases, a set of multiple lines should be used.

Modern VNAs support TRL calibrations and have built-in procedures that ask the operator to connect the various standards, and then calculate the calibration coefficients.

SPECTRUM ANALYZER



Spectrum Analyzer

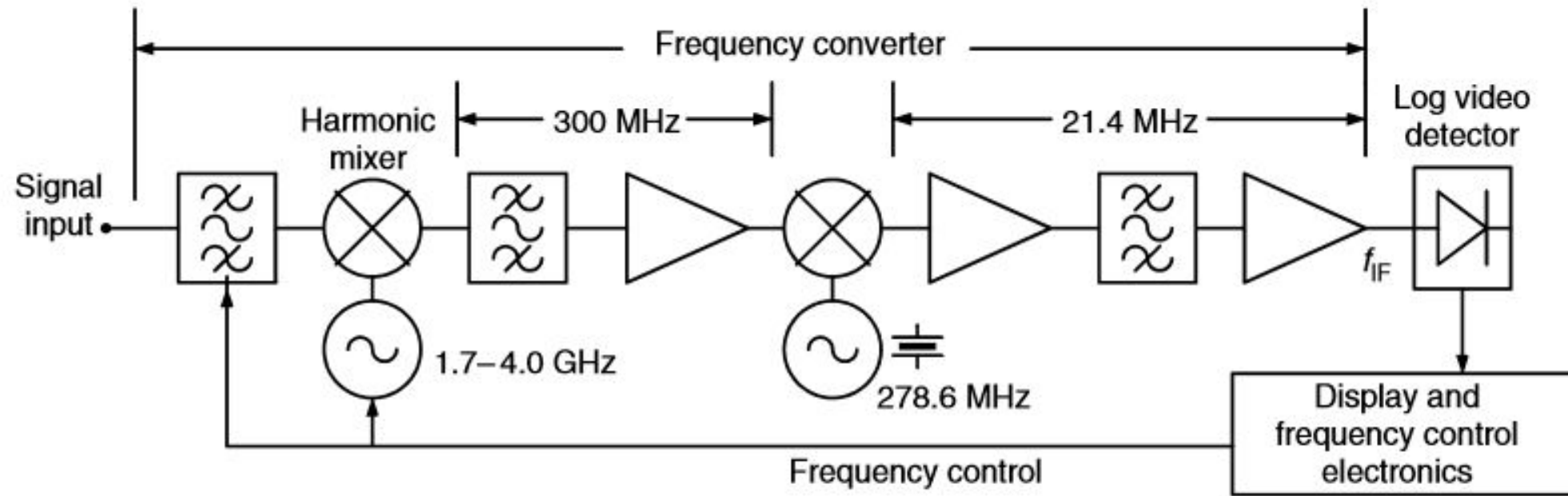


Spectrum Analyzer

The popular manufacturers of RF Spectrum Analyzer are Signal Hound, National Instruments, Anritsu, Rohde & Schwarz, Tektronix, SIGLENT, VIAVI Solutions

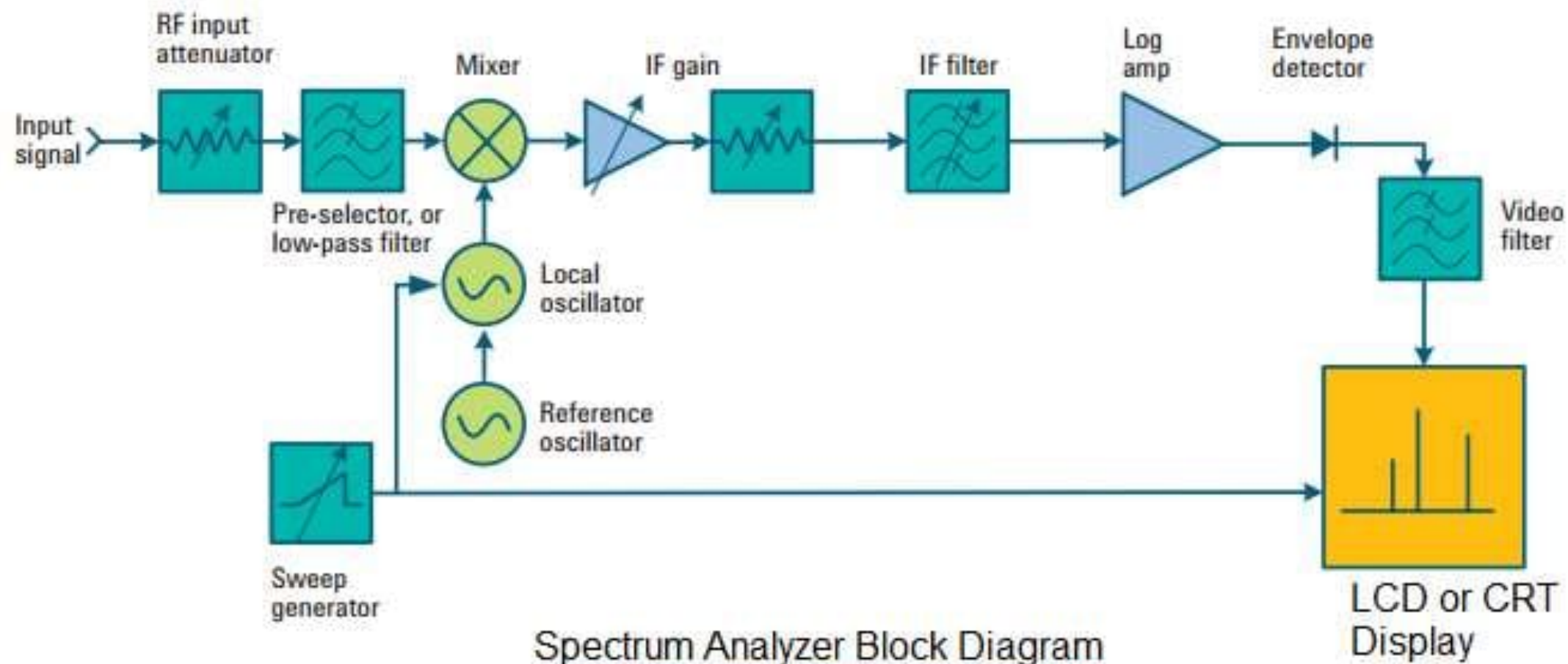
About Spectrum Analyzer

- Spectrum Analyzer is just a receiver which displays the signal fed to its RF input port from any RF transmitting device through cable or with antenna.
- Modern real time spectrum analyzers even decode and display complex broadband RF signals emitted/acquired using various wireless devices which for example include WLAN, WiMAX, GSM, Zigbee, LTE and more.
- Spectrum analyzer do not have source incorporated.
- Spectrum analyzer is used to measure carrier power level, harmonics, spurious, sidebands, phase noise and more.
- The spectrum analyzers usually will have wider range IF bandwidths compare to most of the network analyzers.



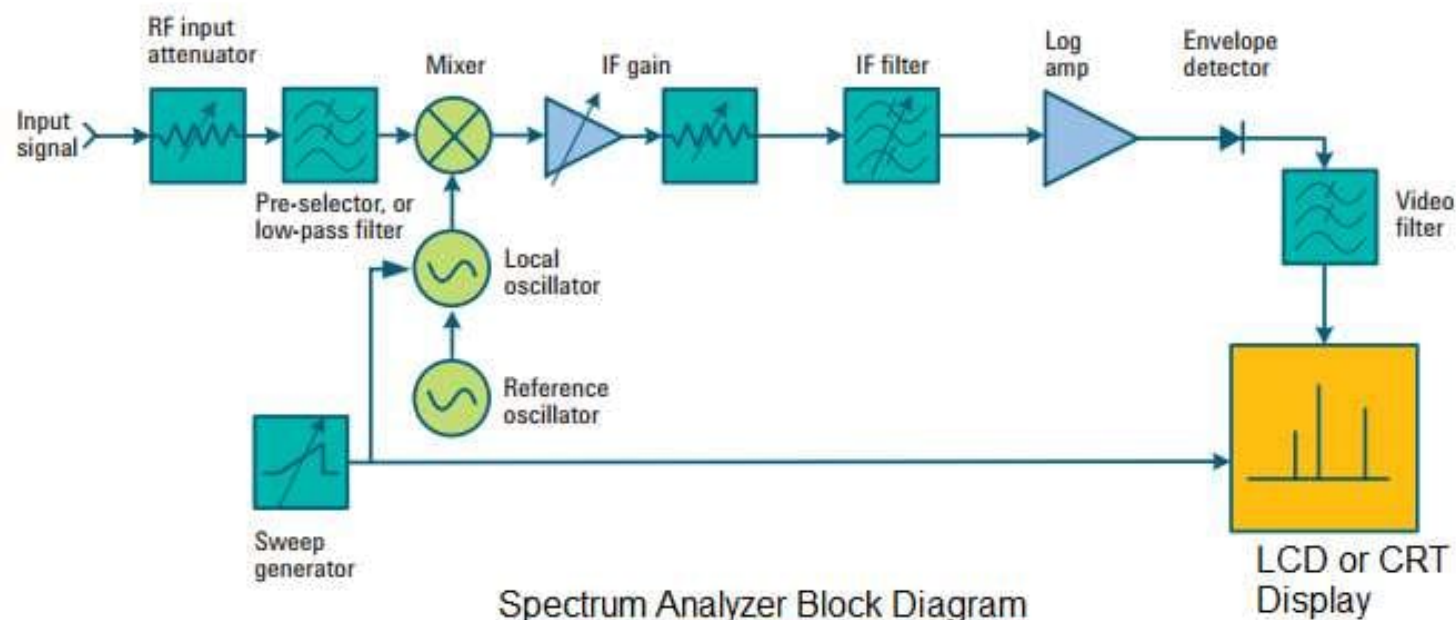
Oscillator harmonic number(n)	Effective oscillator frequency (GHz)	Tuned RF frequency (GHz)
1	1.7–4.0	2.0–4.3
2	3.4–8.0	3.7–8.3
3	5.1–12.0	5.4–12.4
4	6.8–14.0	7.1–14.3
5	8.5–20.0	8.8–20.3

Simplified block diagram of a microwave spectrum analyzer.



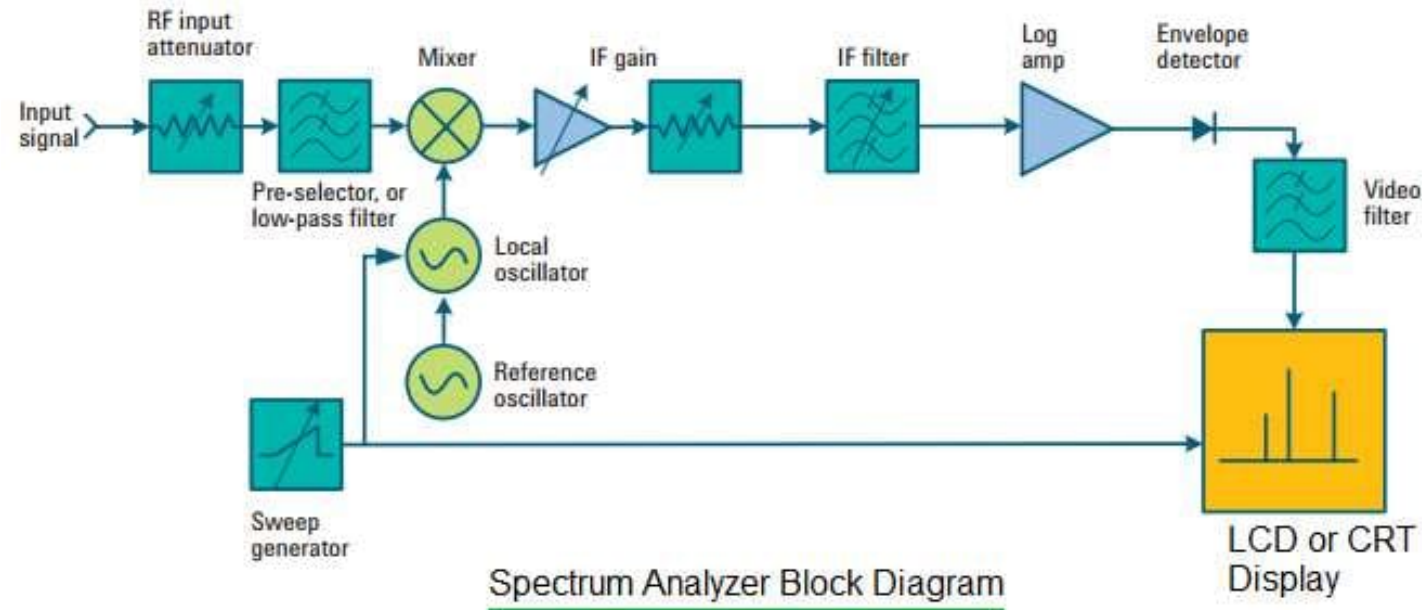
Basic operation of a spectrum analyzer....

- The basic operation starts by connecting the signal to be measured to the input attenuator. The value of this attenuator will have to be changed accordingly by the operator, in order to guarantee that the signal will not drive the input blocks into compression, creating distortion, or even damage.
- The signal after traversing the attenuator will be down/up-converted to the IF frequency by the mixer, which will select which frequency will be sampled by using the local oscillator frequency.
- This IF signal is then filtered out with the IF filter. The bandwidth of this filter controls the resolution bandwidth.



Basic operation of a spectrum analyzer....

- The output signal of this IF filter will then be fed to the envelope detector and the power within this band is measured. The signal is then filtered out by a video filter and displayed in a display, either analog or digital.
- The frequency to be selected for sampling is determined by a swept generator that controls (in the analog SA) the x-axis of the display, which is synchronized with the y-axis where the corresponding power arising from the video filter is plotted. In modern displays this is done digitally using a DSP.
- The speed of the swept generator is actually the sweep time. This should be controlled in order to allow the signal to settle before moving to the next frequency.



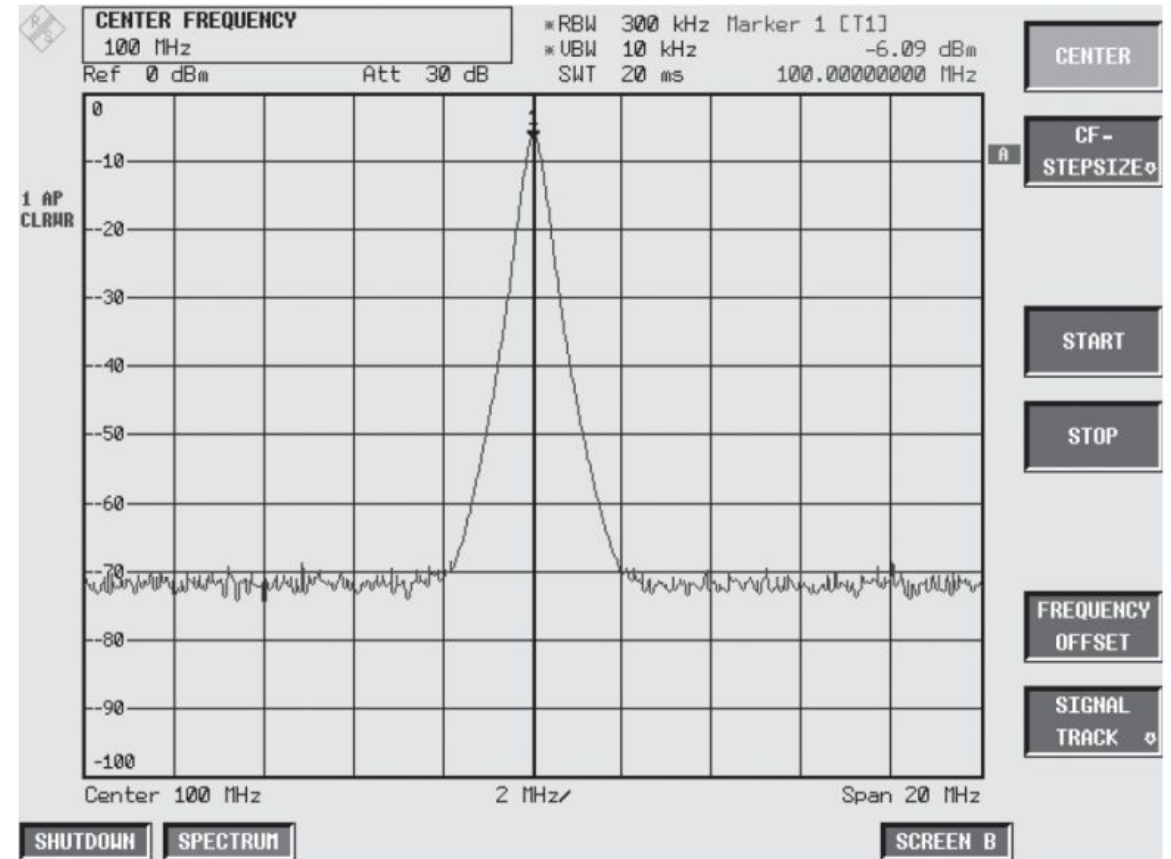
Specifications of a spectrum analyzer

Frequency related characteristics

- Span
- Resolution bandwidth
- Sweep time

SPAN

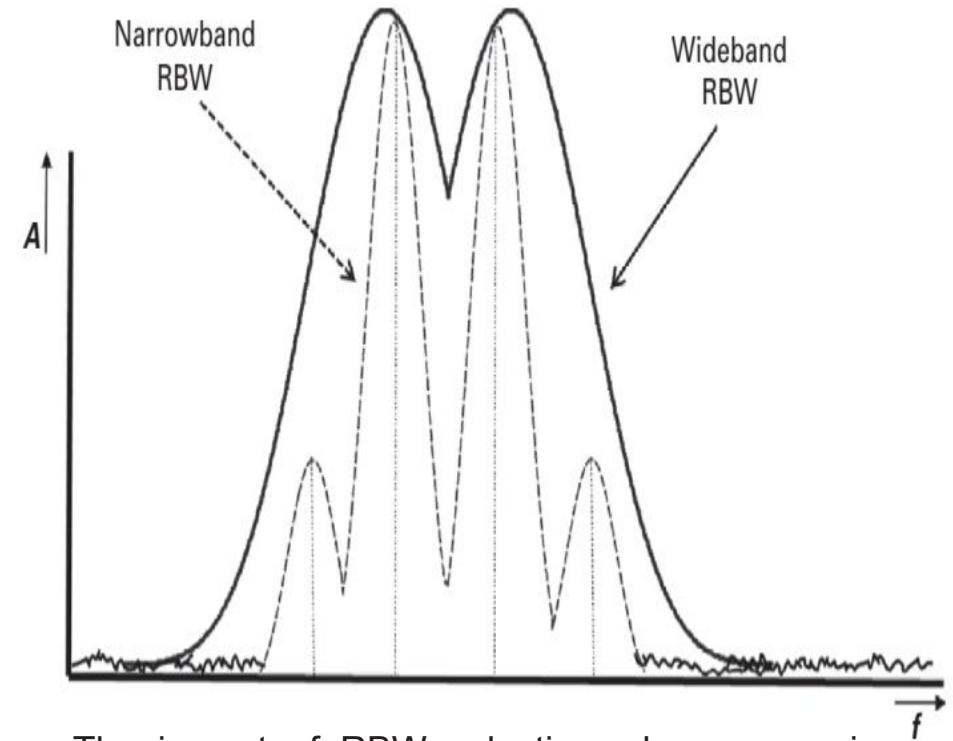
- The span is the frequency range over which the spectrum is to be measured. In most spectrum analyzers we can select the frequency range in two ways: either by selecting the start and stop frequency, or by selecting the central frequency and span.
- In order to settle a span value, the SA instrumentation should control the sawtooth-wave generator, so that it covers the frequency range that is selected, and the RF filter should be able to guarantee that the selected frequency window is visible.



The spectrum-analyzer span and a typical user display

RESOLUTION BANDWIDTH

- This bandwidth is selected by tuning the IF filter, and thus by selecting the amount of power that the envelope detector will measure.
- Resolution is very important, since it allows the SA to distinguish two different frequencies that can be close together, where two different RBWs are selected for evaluating the same signal.
- When a wide RBW is selected, the SA trace becomes so wide that the upper or lower sidebands are measured as if they were the main signal, masking in that respect nearby signals.
- Selecting a narrower RBW allows one to increase the resolution, and therefore to further separate the signals, allowing the identification of each one individually. That is what is meant by selectivity.
- Selectivity is thus the capability of the SA to distinguish between two or more signals with different frequencies. For instance, some manufacturers define selectivity as the ratio of the 60-dB to 3-dB filter bandwidths.



The impact of RBW selection when measuring several close signals in the frequency domain.

SWEEP TIME (ST)

- By selecting the span of the SA, we are determining the settings of the sawtooth-wave generator that controls the tuning of the local oscillator.
- We must also select the sweep time of this generator, which is the speed at which the frequencies will be covered.
- Owing to the fact that the IF (RBW) filter has a settling time, since it is a band-limited filter, the selection of the sweep time is crucial to a good selection of tuned frequencies in the SA. If we had no time delay in the filter, then the sweep time could be as small as we want, since each frequency point would be measured instantaneously; unfortunately, due to the limitations of the filter, we should select the maximum allowable sweep time in order to reduce measurement errors.
- It should also be noticed that narrow RBWs lead to longer sweep times, and wider RBWs to lower sweep times.

$$Time\ in\ IF = \frac{RBW}{SPAN/ST} = \frac{RBW \times ST}{SPAN}$$

Amplitude-related characteristics

- Noise Floor
- Maximum Power
- Dynamic Range

Noise Floor

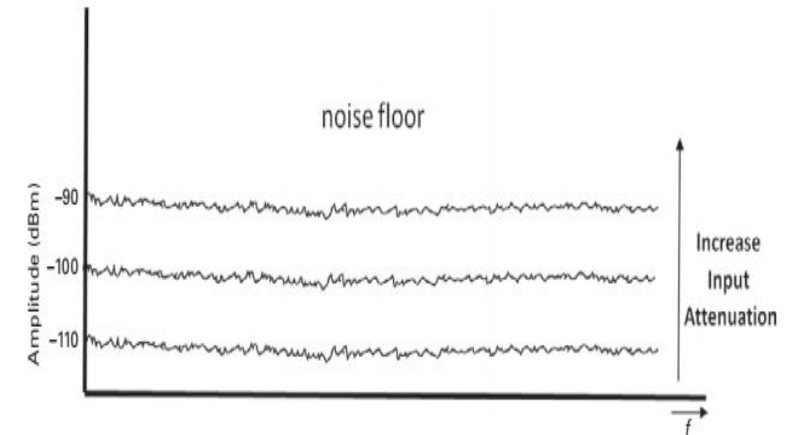
In a spectrum analyzer one of the most important characteristics to know is the minimum signal level that can actually be measured. This is typically identified by a figure of merit called sensitivity.

$$\text{Sensitivity of a receiver}(S_i) = k_B \cdot T \cdot B \cdot F \cdot (SNR)$$

In a spectrum analyzer, the noise floor, N_{floor} , is actually more important than the sensitivity.

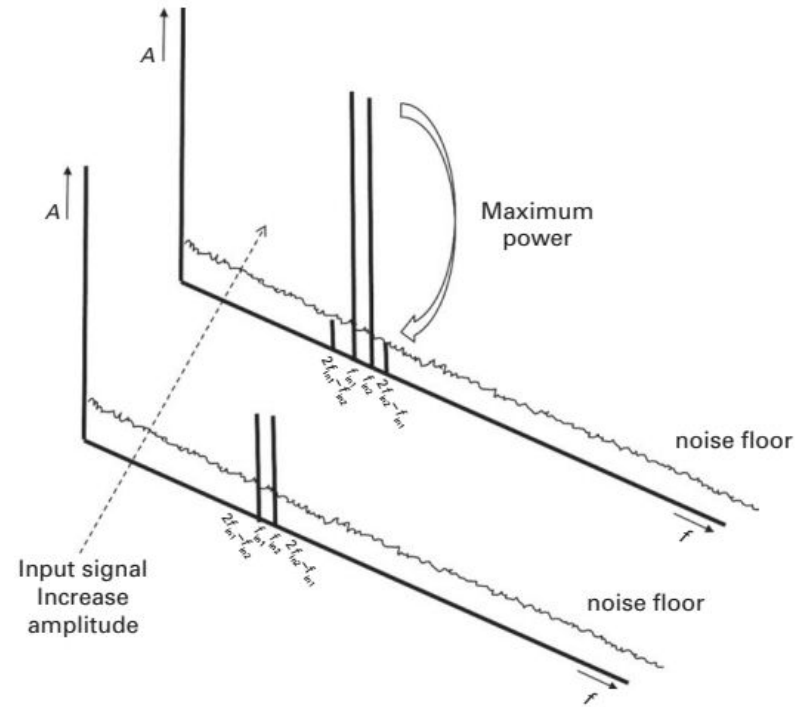
The noise floor is the power level at which the noise is present.

$$N_{floor} = k_B T B F$$



The increase of the SA noise floor with increasing input attenuation.

Maximum Input Power



The maximum admissible input power when the intermodulation distortion starts to appear with a level similar to the noise floor.

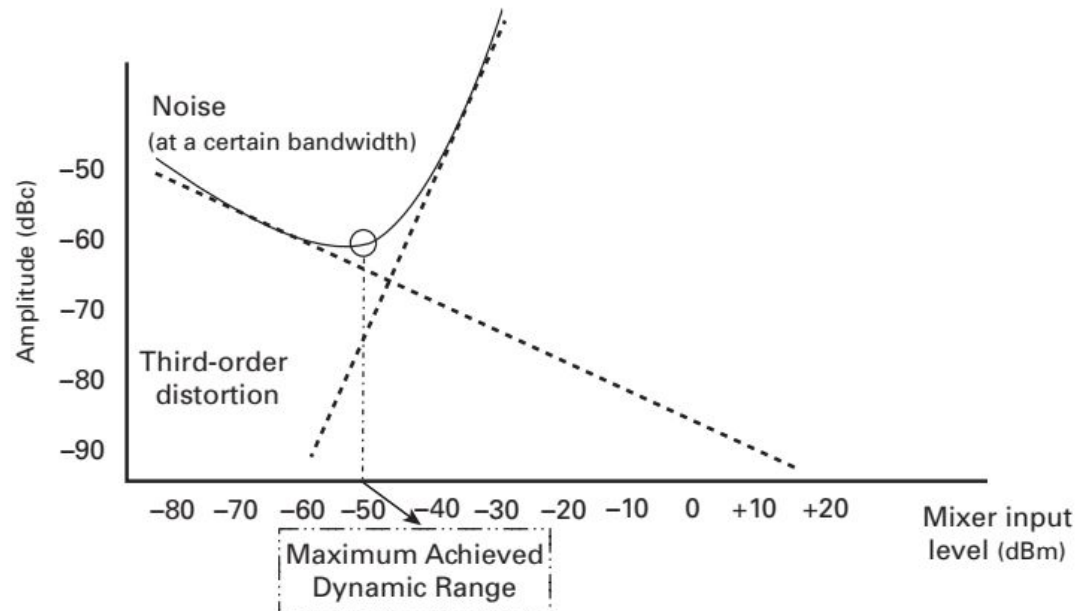
As we increase the input power, the noise level moves further away, but the distortion starts to rise, and the difference between the signal level and the distortion level decreases. This will lead us to an optimum input power that corresponds to the intersection of a distortion curve and the noise characteristic.

Dynamic Range

- The dynamic range is the difference between the maximum power that a circuit or system can handle and the minimum power determined by the noise floor.
- It should be stated that the instrument's dynamic range, within which we assume that the measured signal is free of interference, and the display dynamic range are normally different quantities.

$$\begin{aligned} \text{DR}_{\text{max}} &= \frac{n-1}{n} (\text{IP}_{n_{\text{in}}} - N_{\text{level}}) \\ &= \frac{n-1}{n} (\text{IP}_{n_{\text{in}}} - 10 \log(k_B T) - 10 \log(\text{RBW}) - \text{NF}) \end{aligned}$$

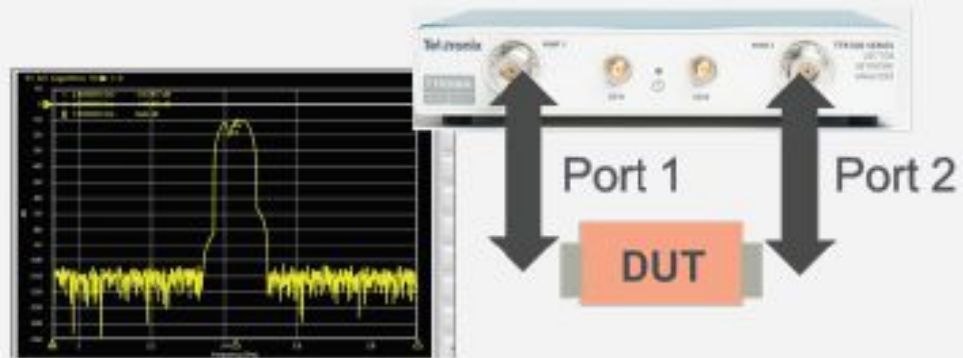
where DR_{max} is the maximum dynamic range, $\text{IP}_{n_{\text{in}}}$ is the intercept point of n th order (when the input attenuation is set to 0 dB), N_{level} is the noise level, n is the intermodulation order, RBW is the resolution bandwidth, and NF is the system's noise figure.



The optimum dynamic-range operation point for a specific application.

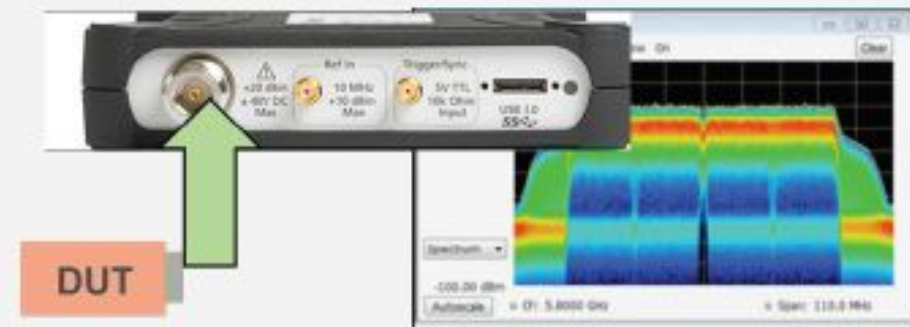
Difference between Network Analyzer and Spectrum Analyzer

Network Analyzer



- Contains a source and receiver(s)
- Uses known stimulus
- 2 channels or more
- Ratioed measurements
- Offers advanced calibration, more accurate
- Limited to analog and pulsed signals

Spectrum Analyzer



- Contains a receiver
- Measures unknown signals
- Single channel
- No ratioed measurements
- Limited calibration capability, less accuracy
- Works well with digitally modulated signals